

Lesson 10 Integrality and Rationality properties of root spaces

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Let L be a semisimple Lie algebra and $L = L_0 + \sum_{\alpha \in H^*} L_\alpha$ its root space decomposition.

Let $\alpha \in \Delta$ ($0 \neq \alpha \in H^*$ s.t. $L_\alpha \neq 0$)

L can be seen as a sl_2 ($\text{span} \{X_\alpha, Y_\alpha, H_\alpha\}$) - module by adjoint representation.

① Consider $\sum_{\alpha \in F} L_\alpha \Rightarrow$ an sl_2 -submodule

L_α belongs to the eigenvalue $\alpha(H_\alpha) = 2c$ of H_α . By classification of irreducible sl_2 -modules, $2c \in \mathbb{Z} \Rightarrow c \in \frac{1}{2}\mathbb{Z}$.

Note that any irreducible sl_2 -modules with even weights has weight 0 $\Rightarrow$ any such sl_2 -submodule S of $\sum_{\alpha \in F} L_\alpha$ has nonempty intersection with H . Pick $0 \neq h \in H \cap S$

If $h \in \ker \alpha$, then Fh is a submodule $\Rightarrow S = Fh$.

If $h \notin \ker \alpha$, then $[h, X_\alpha] \neq 0$, i.e. $X_\alpha \in S$; similarly, $Y_\alpha \in S$, then $[X_\alpha, Y_\alpha] = H_\alpha \in S$, i.e. $\text{span}(X_\alpha, Y_\alpha, H_\alpha) \subset S \Rightarrow S = \text{span}(X_\alpha, Y_\alpha, H_\alpha) = sl_2$

Thus the sum of irreducible submodules with even weights is $H + sl_2$, which yields that $\frac{1}{2}$ is not an eigenvalue, i.e. $c \neq \frac{1}{2} \Rightarrow 2\alpha \notin \Delta$. In particular $\frac{1}{2}\alpha \notin \Delta$ (otherwise $\alpha \notin \Delta$, a contradiction)

But then $\sum_{\alpha \in F} L_\alpha = H + sl_2$ (1 is not an eigenvalue $\Rightarrow$ there exists no irreducible submodules with odd weights).

$\Rightarrow F\Delta \cap \Delta = \{-\alpha, \alpha\}$ and $\dim_F L_\alpha = 1 \quad \forall \alpha \in \Delta$

② Consider $\sum_{i \in \mathbb{Z}} L_{\beta+i\alpha} \Rightarrow$ an sl_2 -module

$L_{\beta+i\alpha}$ belongs to the eigenvalue $\langle \beta+i\alpha, H_\alpha \rangle = \frac{2(\alpha|\beta)}{(\alpha|\alpha)} + 2i \in \mathbb{Z} \Rightarrow \frac{2(\alpha|\beta)}{(\alpha|\alpha)} \in \mathbb{Z}$

Note that all eigenvalues have the same parity and each weight space is 1-dimensional $\Rightarrow$

$\sum_{i \in \mathbb{Z}} L_{\beta+i\alpha}$ is irreducible

Thus if $\alpha + \beta \in \Delta$, $L_{\beta+\alpha} = [L_\beta, L_\alpha]$

And $\exists q, p \in \mathbb{N}$, s.t. the string of $\beta+i\alpha$ s.t. $\beta+i\alpha \in \Delta$ is

$\beta - q\alpha, \dots, \beta - \alpha, \beta, \beta + \alpha, \dots, \beta + p\alpha$
where $q - p = \frac{2(\alpha|\beta)}{(\alpha|\alpha)}$

Def reflection

Consider $S_\alpha: X \mapsto X - \frac{2(X|\alpha)}{(\alpha|\alpha)} \alpha$

Def reflection

Consider $S_\alpha: x \mapsto x - \frac{2(x|\alpha)}{(\alpha|\alpha)} \alpha$

Note that S_α sends α to $-\alpha$

x to x

$\forall x \in \{\beta \mid (\beta|\alpha) = 0\}$ a hyperplane (subspace of codimen 1)

Lemma $\forall \alpha \in \Delta, S_\alpha(\Delta) = \Delta$

pf: Since S_α is a bijection, it suffices to show $S_\alpha(\Delta) \subset \Delta$

i.e. $\forall \beta \in \Delta, \beta - \frac{2(\beta|\alpha)}{(\alpha|\alpha)} \alpha \in \Delta \Leftrightarrow -q \leq \frac{2(\beta|\alpha)}{(\alpha|\alpha)} \leq p \Leftrightarrow -q \leq p-q \leq p$